

```
kavya@kavya-VirtualBox:~$ ping google.com
```

```
PING google.com (142.251.221.174) 56(84) bytes of data.
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=1 ttl=255 time=47.7 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=2 ttl=255 time=54.2 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=3 ttl=255 time=54.3 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=4 ttl=255 time=51.8 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=5 ttl=255 time=61.5 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=6 ttl=255 time=65.3 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=7 ttl=255 time=44.5 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=8 ttl=255 time=51.5 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=9 ttl=255 time=48.8 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=10 ttl=255 time=68.1 ms
```

```
64 bytes from pnmaaa-au-in-f14.1e100.net (142.251.221.174): icmp_seq=11 ttl=255 time=44.8 ms
```

```

General statistics:
    total time:                10.0004s
    total number of events:    60408024

Latency (ms):
    min:                      0.00
    avg:                      0.00
    max:                      2.24
    95th percentile:         0.00
    sum:                      4054.31

Threads fairness:
    events (avg/stddev):       60408024.0000/0.00
    execution time (avg/stddev): 4.0543/0.00

kavya@kavya-VirtualBox:~$ man sysbench

kavya@kavya-VirtualBox:~$ stress-ng --version
stress-ng, version 0.17.06 (gcc 13.2.0, x86_64 Linux 6.14.0
36-generic) 🖥️🔥

kavya@kavya-VirtualBox:~$ stress-ng --cpu 0 --timeout 60s
stress-ng: info:  [5527] setting to a 1 min, 0 secs run per
stressor
stress-ng: info:  [5527] dispatching hogs: 7 cpu
stress-ng: info:  [5527] skipped: 0
stress-ng: info:  [5527] passed: 7: cpu (7)
stress-ng: info:  [5527] failed: 0
stress-ng: info:  [5527] metrics untrustworthy: 0
stress-ng: info:  [5527] successful run completed in 1 min,
0.42 secs

kavya@kavya-VirtualBox:~$ █

```

```

top - 17:16:40 up 37 min,  1 user,  load average: 0.44, 0.
Tasks: 264 total,  1 running, 263 sleeping,  0 stopped,
%Cpu(s):  0.2 us,  0.4 sy,  0.0 ni, 99.3 id,  0.0 wa,  0.0
MiB Mem :  7163.8 total,  4138.3 free,  1192.1 used,
MiB Swap:  4096.0 total,  4096.0 free,    0.0 used.

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU
2451	kavya	20	0	5630856	416972	147560	S	7.4
3172	kavya	20	0	581272	59300	45044	S	1.2
5566	kavya	20	0	14516	5900	3668	R	0.6
74	root	20	0	0	0	0	I	0.3
474	systemd+	20	0	17560	7660	6752	S	0.3
3474	root	20	0	0	0	0	I	0.3
5542	root	20	0	0	0	0	I	0.3
1	root	20	0	23196	14444	9524	S	0.0
2	root	20	0	0	0	0	S	0.0
3	root	20	0	0	0	0	S	0.0
4	root	0	-20	0	0	0	I	0.0
5	root	0	-20	0	0	0	I	0.0
6	root	0	-20	0	0	0	I	0.0
7	root	0	-20	0	0	0	I	0.0
8	root	0	-20	0	0	0	I	0.0
12	root	20	0	0	0	0	I	0.0
13	root	0	-20	0	0	0	I	0.0
14	root	20	0	0	0	0	I	0.0
15	root	20	0	0	0	0	I	0.0
16	root	20	0	0	0	0	I	0.0
17	root	20	0	0	0	0	S	0.0
18	root	20	0	0	0	0	I	0.0
19	root	20	0	0	0	0	S	0.0
20	root	20	0	0	0	0	S	0.0
21	root	rt	0	0	0	0	S	0.0

kavya@kavya-VirtualBox:~\$ iperf3 -c 10.242.51.190

Connecting to host 10.242.51.190, port 5201

[5] local 10.242.51.190 port 41784 connected to 10.242.51.190 port 5201

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	6.86 GBytes	58.9 Gbits/sec	0	2.94 MBytes
[5]	1.00-2.00	sec	6.87 GBytes	59.0 Gbits/sec	0	3.56 MBytes
[5]	2.00-3.00	sec	7.10 GBytes	61.0 Gbits/sec	0	3.75 MBytes
[5]	3.00-4.00	sec	7.16 GBytes	61.5 Gbits/sec	0	4.12 MBytes
[5]	4.00-5.00	sec	7.23 GBytes	62.1 Gbits/sec	0	4.12 MBytes
[5]	5.00-6.00	sec	7.71 GBytes	66.1 Gbits/sec	1	4.12 MBytes
[5]	6.00-7.00	sec	7.37 GBytes	63.5 Gbits/sec	0	4.12 MBytes
[5]	7.00-8.00	sec	7.37 GBytes	63.2 Gbits/sec	0	4.12 MBytes
[5]	8.00-9.00	sec	7.47 GBytes	64.3 Gbits/sec	0	4.12 MBytes
[5]	9.00-10.00	sec	7.60 GBytes	65.3 Gbits/sec	0	4.12 MBytes

[ID]	Interval		Transfer	Bitrate	Retr
[5]	0.00-10.00	sec	72.9 GBytes	62.6 Gbits/sec	1
[5]	0.00-10.00	sec	72.9 GBytes	62.6 Gbits/sec	

sender
receiver

iperf Done.

kavya@kavya-VirtualBox:~\$

```
kavya@kavya-VirtualBox: ~  
kavya@kavya-VirtualBox:~$ fio --name=write --rw=write --size=250M  
write: (g=0): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1  
fio-3.36  
Starting 1 process  
write: Laying out IO file (1 file / 250MiB)  
  
write: (groupid=0, jobs=1): err= 0: pid=6102: Mon Dec  8 17:25:21 2025  
  write: IOPS=229k, BW=893MiB/s (936MB/s)(250MiB/280msec); 0 zone resets  
    clat (usec): min=2, max=12868, avg= 4.13, stdev=51.82  
    lat (usec): min=2, max=12868, avg= 4.17, stdev=51.83  
    clat percentiles (usec):  
      |  1.00th=[  3],  5.00th=[  3], 10.00th=[  3], 20.00th=[  3],  
      | 30.00th=[  3], 40.00th=[  3], 50.00th=[  3], 60.00th=[  3],  
      | 70.00th=[  4], 80.00th=[  4], 90.00th=[  5], 95.00th=[  6],  
      | 99.00th=[ 22], 99.50th=[ 43], 99.90th=[ 169], 99.95th=[ 221],  
      | 99.99th=[ 363]  
    lat (usec)   : 4=86.69%, 10=11.17%, 20=1.05%, 50=0.62%, 100=0.22%  
    lat (usec)   : 250=0.22%, 500=0.03%  
    lat (msec)   : 20=0.01%  
    cpu          : usr=7.17%, sys=91.40%, ctx=0, majf=0, minf=13  
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%  
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%  
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%  
      issued rwts: total=0,64000,0,0 short=0,0,0,0 dropped=0,0,0,0  
      latency    : target=0, window=0, percentile=100.00%, depth=1  
  
Run status group 0 (all jobs):  
  WRITE: bw=893MiB/s (936MB/s), 893MiB/s-893MiB/s (936MB/s-936MB/s), io=250MiB (262MB), run=280-280msec  
  
Disk stats (read/write):  
  sda: ios=20/2, sectors=1032/344, merge=0/41, ticks=13/38, in queue=77, util=15.77%
```

Name

VirtualBox Host-Only Ethernet Adapter

VirtualBox Host-Only Ethernet Adapter #2

Adapter

DHCP Server



Enable Server

Server Address: 192.168.1.2

Server Mask: 255.255.255.0

Lower Address Bound: 192.168.1.3

Upper Address Bound: 192.168.1.254

kavya@kavya-VirtualBox:~\$ iperf3 -c 10.242.51.190

Connecting to host 10.242.51.190, port 5201

[5] local 10.242.51.190 port 41784 connected to 10.242.51.190 port 5201

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	6.86 GBytes	58.9 Gbits/sec	0	2.94 MBytes
[5]	1.00-2.00	sec	6.87 GBytes	59.0 Gbits/sec	0	3.56 MBytes
[5]	2.00-3.00	sec	7.10 GBytes	61.0 Gbits/sec	0	3.75 MBytes
[5]	3.00-4.00	sec	7.16 GBytes	61.5 Gbits/sec	0	4.12 MBytes
[5]	4.00-5.00	sec	7.23 GBytes	62.1 Gbits/sec	0	4.12 MBytes
[5]	5.00-6.00	sec	7.71 GBytes	66.1 Gbits/sec	1	4.12 MBytes
[5]	6.00-7.00	sec	7.37 GBytes	63.5 Gbits/sec	0	4.12 MBytes
[5]	7.00-8.00	sec	7.37 GBytes	63.2 Gbits/sec	0	4.12 MBytes
[5]	8.00-9.00	sec	7.47 GBytes	64.3 Gbits/sec	0	4.12 MBytes
[5]	9.00-10.00	sec	7.60 GBytes	65.3 Gbits/sec	0	4.12 MBytes

[ID]	Interval		Transfer	Bitrate	Retr
[5]	0.00-10.00	sec	72.9 GBytes	62.6 Gbits/sec	1
[5]	0.00-10.00	sec	72.9 GBytes	62.6 Gbits/sec	

sender
receiver

iperf Done.

kavya@kavya-VirtualBox:~\$

```
kavya@kavya-VirtualBox: ~  
kavya@kavya-VirtualBox:~$ fio --name=write --rw=write --size=250M  
write: (g=0): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1  
fio-3.36  
Starting 1 process  
write: Laying out IO file (1 file / 250MiB)  
  
write: (groupid=0, jobs=1): err= 0: pid=6102: Mon Dec  8 17:25:21 2025  
  write: IOPS=229k, BW=893MiB/s (936MB/s)(250MiB/280msec); 0 zone resets  
    clat (usec): min=2, max=12868, avg= 4.13, stdev=51.82  
    lat (usec): min=2, max=12868, avg= 4.17, stdev=51.83  
    clat percentiles (usec):  
      |  1.00th=[  3],  5.00th=[  3], 10.00th=[  3], 20.00th=[  3],  
      | 30.00th=[  3], 40.00th=[  3], 50.00th=[  3], 60.00th=[  3],  
      | 70.00th=[  4], 80.00th=[  4], 90.00th=[  5], 95.00th=[  6],  
      | 99.00th=[ 22], 99.50th=[ 43], 99.90th=[ 169], 99.95th=[ 221],  
      | 99.99th=[ 363]  
    lat (usec)   : 4=86.69%, 10=11.17%, 20=1.05%, 50=0.62%, 100=0.22%  
    lat (usec)   : 250=0.22%, 500=0.03%  
    lat (msec)   : 20=0.01%  
    cpu          : usr=7.17%, sys=91.40%, ctx=0, majf=0, minf=13  
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%  
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%  
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%  
      issued rwts: total=0,64000,0,0 short=0,0,0,0 dropped=0,0,0,0  
      latency    : target=0, window=0, percentile=100.00%, depth=1  
  
Run status group 0 (all jobs):  
  WRITE: bw=893MiB/s (936MB/s), 893MiB/s-893MiB/s (936MB/s-936MB/s), io=250MiB (262MB), run=280-280msec  
  
Disk stats (read/write):  
  sda: ios=20/2  sectors=1032/344  merge=0/41  ticks=13/38  in queue=77  util=15.77%
```

```
kavya@kavya-VirtualBox:~$ fio --name=write --rw=write --size=250M
write: (g=0): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
fio-3.36
Starting 1 process
write: Laying out IO file (1 file / 250MiB)

write: (groupid=0, jobs=1): err= 0: pid=6102: Mon Dec  8 17:25:21 2025
  write: IOPS=229k, BW=893MiB/s (936MB/s)(250MiB/280msec); 0 zone resets
    clat (usec): min=2, max=12868, avg= 4.13, stdev=51.82
    lat (usec): min=2, max=12868, avg= 4.17, stdev=51.83
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      |  1.00th=[  3],  5.00th=[  3], 10.00th=[  3], 20.00th=[  3],
      | 30.00th=[  3], 40.00th=[  3], 50.00th=[  3], 60.00th=[  3],
      | 70.00th=[  4], 80.00th=[  4], 90.00th=[  5], 95.00th=[  6],
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    lat (usec)   : 250=0.22%, 500=0.03%
    lat (msec)   : 20=0.01%
    cpu          : usr=7.17%, sys=91.40%, ctx=0, majf=0, minf=13
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      issued rwts: total=0,64000,0,0 short=0,0,0,0 dropped=0,0,0,0
      latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=893MiB/s (936MB/s), 893MiB/s-893MiB/s (936MB/s-936MB/s), io=250MiB (262MB), run=280-280msec

Disk stats (read/write):
  sda: ios=20/2, sectors=1032/344, merge=0/41, ticks=13/38, in_queue=77, util=15.77%
kavya@kavya-VirtualBox:~$
```

```

General statistics:
    total time:                10.0004s
    total number of events:    60408024

Latency (ms):
    min:                       0.00
    avg:                       0.00
    max:                       2.24
    95th percentile:         0.00
    sum:                       4054.31

Threads fairness:
    events (avg/stddev):       60408024.0000/0.00
    execution time (avg/stddev): 4.0543/0.00

kavya@kavya-VirtualBox:~$ man sysbench

kavya@kavya-VirtualBox:~$ stress-ng --version
stress-ng, version 0.17.06 (gcc 13.2.0, x86_64 Linux 6.14.0
36-generic) 🖥️🔥

kavya@kavya-VirtualBox:~$ stress-ng --cpu 0 --timeout 60s
stress-ng: info:  [5527] setting to a 1 min, 0 secs run per
stressor
stress-ng: info:  [5527] dispatching hogs: 7 cpu
stress-ng: info:  [5527] skipped: 0
stress-ng: info:  [5527] passed: 7: cpu (7)
stress-ng: info:  [5527] failed: 0
stress-ng: info:  [5527] metrics untrustworthy: 0
stress-ng: info:  [5527] successful run completed in 1 min,
0.42 secs

kavya@kavya-VirtualBox:~$ █

```

```

top - 17:16:40 up 37 min,  1 user,  load average: 0.44, 0.
Tasks: 264 total,   1 running, 263 sleeping,   0 stopped,
%Cpu(s):  0.2 us,  0.4 sy,  0.0 ni, 99.3 id,  0.0 wa,  0.0
MiB Mem :  7163.8 total,  4138.3 free,  1192.1 used,
MiB Swap: 4096.0 total,  4096.0 free,    0.0 used.

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU
2451	kavya	20	0	5630856	416972	147560	S	7.4
3172	kavya	20	0	581272	59300	45044	S	1.2
5566	kavya	20	0	14516	5900	3668	R	0.6
74	root	20	0	0	0	0	I	0.3
474	systemd+	20	0	17560	7660	6752	S	0.3
3474	root	20	0	0	0	0	I	0.3
5542	root	20	0	0	0	0	I	0.3
1	root	20	0	23196	14444	9524	S	0.0
2	root	20	0	0	0	0	S	0.0
3	root	20	0	0	0	0	S	0.0
4	root	0	-20	0	0	0	I	0.0
5	root	0	-20	0	0	0	I	0.0
6	root	0	-20	0	0	0	I	0.0
7	root	0	-20	0	0	0	I	0.0
8	root	0	-20	0	0	0	I	0.0
12	root	20	0	0	0	0	I	0.0
13	root	0	-20	0	0	0	I	0.0
14	root	20	0	0	0	0	I	0.0
15	root	20	0	0	0	0	I	0.0
16	root	20	0	0	0	0	I	0.0
17	root	20	0	0	0	0	S	0.0
18	root	20	0	0	0	0	I	0.0
19	root	20	0	0	0	0	S	0.0
20	root	20	0	0	0	0	S	0.0
21	root	rt	0	0	0	0	S	0.0

General statistics:

total time:	10.0004s
total number of events:	60408024

Latency (ms):

min:	0.00
avg:	0.00
max:	2.24
95th percentile:	0.00
sum:	4054.31

Threads fairness:

events (avg/stddev):	60408024.0000/0.00
execution time (avg/stddev):	4.0543/0.00

kavya@kavya-VirtualBox:~\$ man sysbench

kavya@kavya-VirtualBox:~\$ stress-ng --version

stress-ng, version 0.17.06 (gcc 13.2.0, x86_64 Linux 6.14.0-36-generic) 🖥️🔥

kavya@kavya-VirtualBox:~\$ stress-ng --cpu 0 --timeout 60s

stress-ng: info: [5527] setting to a 1 min, 0 secs run per stressor

stress-ng: info: [5527] dispatching hogs: 7 cpu

stress-ng: info: [5527] skipped: 0

stress-ng: info: [5527] passed: 7: cpu (7)

stress-ng: info: [5527] failed: 0

stress-ng: info: [5527] metrics untrustworthy: 0

stress-ng: info: [5527] successful run completed in 1 min, 0.42 secs

kavya@kavya-VirtualBox:~\$

```
events (avg/stddev):      60408024.0000/0.00  
execution time (avg/stddev):  4.0543/0.00
```

```
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```

```
kavya@kavya-VirtualBox:~$ stress-ng --version
```

```
stress-ng, version 0.17.06 (gcc 13.2.0, x86_64 Linux 6.14.0-36-generic) 🖥️🔥
```

```
kavya@kavya-VirtualBox:~$ stress-ng --cpu 0 --timeout 60s
```

```
stress-ng: info:  [5527] setting to a 1 min, 0 secs run per stressor
```

```
stress-ng: info:  [5527] dispatching hogs: 7 cpu
```

```
stress-ng: info:  [5527] skipped: 0
```

```
stress-ng: info:  [5527] passed: 7: cpu (7)
```

```
stress-ng: info:  [5527] failed: 0
```

```
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```
stress-ng: info:  [5527] successful run completed in 1 min, 0.42 secs
```

```
kavya@kavya-VirtualBox:~$ █
```

Host-only Networks NAT Networks Cloud Networks

Name

VirtualBox Host-Only Ethernet Adapter

VirtualBox Host-Only Ethernet Adapter #2

Adapter DHCP Server

☐ Configure Adapter Automatically

☒ Configure Adapter Manually

IPv4 Address: 192.168.1.1

IPv4 Network Mask: 255.255.255.0

IPv6 Address: fe80::881:685f:8094:bda4

IPv6 Prefix Length: 64

Name

VirtualBox Host-Only Ethernet Adapter

VirtualBox Host-Only Ethernet Adapter #2

Adapter

DHCP Server



Enable Server

Server Address: 192.168.1.2

Server Mask: 255.255.255.0

Lower Address Bound: 192.168.1.3

Upper Address Bound: 192.168.1.254